

TRAINING PROJECT REPORT

DESIGN AND RTL-TO-GDSII IMPLEMENTATION OF A SMART 3-WAY TRAFFIC LIGHT CONTROLLER USING CADENCE TOOLS

Submitted by

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2026

ABSTRACT

This project presents the design and RTL-to-GDSII implementation of a Smart 3-Way Traffic Light Controller using Verilog HDL and Cadence EDA tools. The traffic controller is designed using a Finite State Machine (FSM) approach to manage traffic flow among three different roads by controlling red, yellow, and green light signals.

The design was first modeled and functionally verified at the RTL level using Verilog HDL. After successful verification, the design was synthesized using Cadence Genus to generate a gate-level netlist. Physical design implementation was then carried out using Cadence Innovus, including floorplanning, power planning, placement, clock tree synthesis (CTS), routing, and post-route timing analysis.

The final design successfully achieved timing closure with zero setup and hold violations. Post-route analysis reported a setup slack of 7.643 ns and a hold slack of 0.339 ns. The design also achieved zero DRC violations and zero routing overflow. The total power consumption of the design was measured as 0.027289 mW.

The project demonstrates the complete ASIC design flow from RTL design to physical implementation and provides practical exposure to modern VLSI design methodologies using industry-standard Cadence tools.

INTRODUCTION

Traffic management is an essential part of modern transportation systems. Efficient control of traffic signals helps reduce congestion, improve road safety, and ensure smooth vehicle movement at road intersections. Traffic light controllers are widely used to manage traffic flow by controlling the sequence and duration of red, yellow, and green signals.

In digital system design, traffic light controllers are commonly implemented using Finite State Machines (FSMs), where each state represents a specific traffic condition. FSM-based designs provide a structured and reliable approach for controlling signal transitions and timing operations.

In this project, a Smart 3-Way Traffic Light Controller was designed using Verilog HDL and implemented through the complete ASIC design flow using Cadence EDA tools. The design was functionally verified at the RTL level and then synthesized using Cadence Genus. The synthesized netlist was further implemented using Cadence Innovus, where floorplanning, power planning, placement, clock tree synthesis, routing, and timing analysis were performed.

The objective of this project is to gain practical experience in the RTL-to-GDSII design flow and understand the various stages involved in ASIC implementation using industry-standard VLSI design tools.

Design Specification

The Smart 3-Way Traffic Light Controller is designed to control traffic flow at a three-road intersection. The controller operates using a Finite State Machine (FSM) and generates traffic light signals for three directions, namely Road A, Road B, and Road C.

1. Inputs and Outputs

Inputs

Signal	Width	Description
clk	1 bit	System clock
rst	1 bit	Asynchronous reset

Outputs

Signal	Width	Description
A	3 bits	Traffic signal for Road A
B	3 bits	Traffic signal for Road B
C	3 bits	Traffic signal for Road C

2. Traffic Light Encoding

The traffic light outputs are encoded as follows:

Encoding	Signal
3'b001	Green
3'b010	Yellow
3'b100	Red

3. Timing Parameters

The controller uses fixed timing values for signal transitions.

Parameter	Value
GREEN_TIME	10 clock cycles
YELLOW_TIME	3 clock cycles

4. FSM States

The controller consists of six states:

State	Description
S0	Road A Green
S1	Road A Yellow
S2	Road B Green
S3	Road B Yellow
S4	Road C Green
S5	Road C Yellow

The controller sequentially transitions through these states to provide fair traffic distribution among all three roads.

5. Working Principle

Initially, Road A is assigned the green signal while Roads B and C remain red. After the specified green duration, Road A transitions to yellow before turning red. The same sequence is repeated for Roads B and C. The process continues cyclically, ensuring orderly traffic flow through the intersection.

RTL Design

RTL Verilog Code

```
1 module smart_traffic_3way (
2     input clk,
3     input rst,
4     input ped_req,
5     input emergency,
6
7     output reg [2:0] A,
8     output reg [2:0] B,
9     output reg [2:0] C
10 );
11
12 // Light Encoding
13 // 001 = GREEN
14 // 010 = YELLOW
15 // 100 = RED
16
17 localparam S0 = 4'd0; // A Green
18 localparam S1 = 4'd1; // A Yellow
19 localparam S2 = 4'd2; // B Green
20 localparam S3 = 4'd3; // B Yellow
21 localparam S4 = 4'd4; // C Green
22 localparam S5 = 4'd5; // C Yellow
23 localparam PED = 4'd6; // Pedestrian Crossing
24 localparam EM = 4'b0110; // Emergency Mode
25
26 localparam GREEN_TIME = 20;
27 localparam YELLOW_TIME = 5;
28 localparam PED_TIME = 10;
29
30 reg [3:0] ps, ns;
31 reg [5:0] count;
32
33 //-----
34 // State Register
35 //-----
36 always @(posedge clk or posedge rst) begin
37     if (rst)
38         ps <= S0;
39     else
40         ps <= ns;
41 end
42
43 //-----
44 // Counter
45 //-----
46 always @(posedge clk or posedge rst) begin
47     if (rst)
48         count <= 0;
49     else if (ps != ns)
50         count <= 0;
51     else
52         count <= count + 1;
53 end
54
```

```

55 //-----
56 // Next State Logic
57 //-----
58 always @(*) begin
59
60     ns = ps;
61
62     case(ps)
63
64         S0:
65             if(emergency)
66                 ns = EMG;
67             else if(count >= GREEN_TIME)
68                 ns = S1;
69
70         S1:
71             if(emergency)
72                 ns = EMG;
73             else if(count >= YELLOW_TIME)
74                 ns = (ped_req) ? PED : S2;
75
76         S2:
77             if(emergency)
78                 ns = EMG;
79             else if(count >= GREEN_TIME)
80                 ns = S3;
81
82         S3:
83             if(emergency)
84                 ns = EMG;
85             else if(count >= YELLOW_TIME)
86                 ns = (ped_req) ? PED : S4;
87
88         S4:
89             if(emergency)
90                 ns = EMG;
91             else if(count >= GREEN_TIME)
92                 ns = S5;
93
94         S5:
95             if(emergency)
96                 ns = EMG;
97             else if(count >= YELLOW_TIME)
98                 ns = (ped_req) ? PED : S0;
99
100        PED:
101            if(emergency)
102                ns = EMG;
103            else if(count >= PED_TIME)
104                ns = S0;
105
106        EMG:
107            if(!emergency)
108                ns = S0;
109
110        default:
111            ns = S0;
112

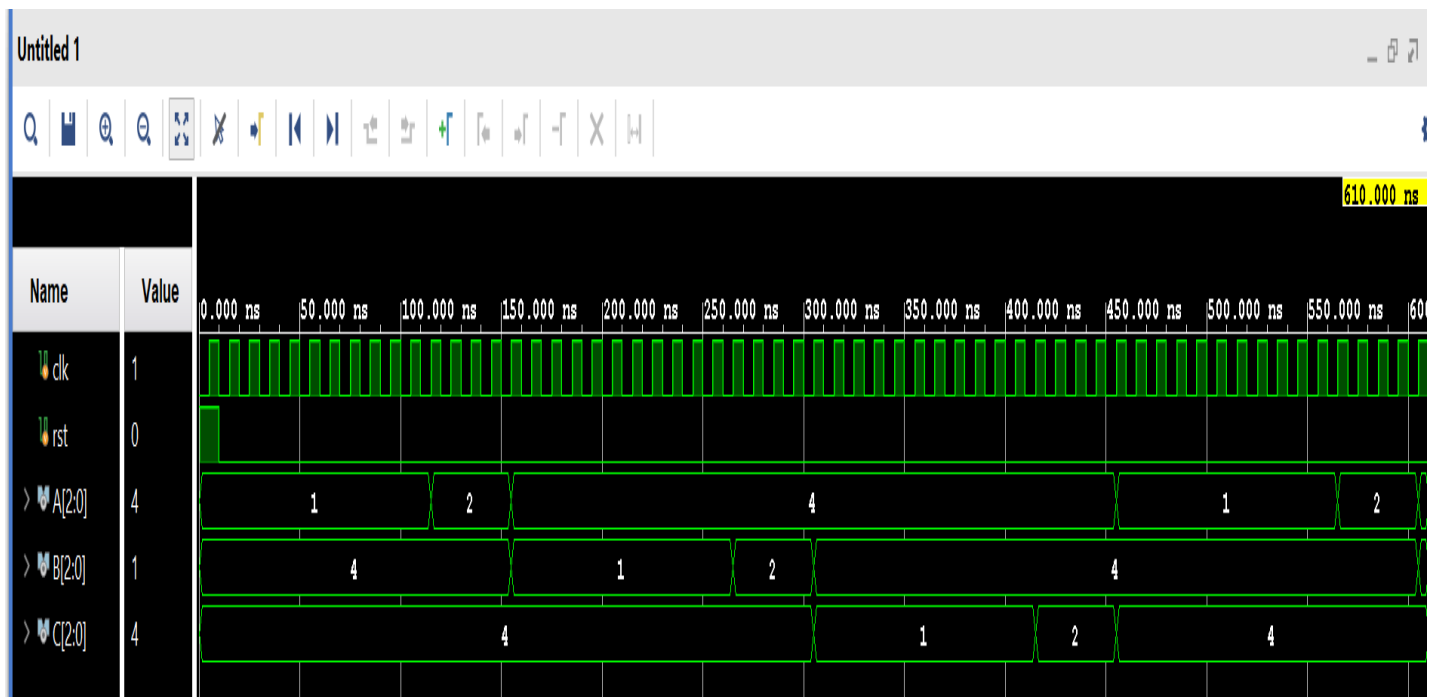
```

```

116 //-----
117 // Output Logic
118 //-----
119 always @(*) begin
120
121     A = 3'b100;
122     B = 3'b100;
123     C = 3'b100;
124
125     case(ps)
126
127     S0: begin
128         A = 3'b001;
129         B = 3'b100;
130         C = 3'b100;
131     end
132
133     S1: begin
134         A = 3'b010;
135         B = 3'b100;
136         C = 3'b100;
137     end
138
139     S2: begin
140         A = 3'b100;
141         B = 3'b001;
142         C = 3'b100;
143     end
144
145     S3: begin
146         A = 3'b100;
147         B = 3'b010;
148         C = 3'b100;
149     end
150
151     S4: begin
152         A = 3'b100;
153         B = 3'b100;
154         C = 3'b001;
155     end
156
157     S5: begin
158         A = 3'b100;
159         B = 3'b100;
160         C = 3'b010;
161     end
162
163     // Pedestrian Crossing
164     PED: begin
165         A = 3'b100;
166         B = 3'b100;
167         C = 3'b100;
168     end
169
170     // Emergency Vehicle Priority
171     // Road A gets Green
172     EMG: begin
173         A = 3'b001;
174         B = 3'b100;
175         C = 3'b100;
176     end
177
178

```


RTL Simulation Waveform

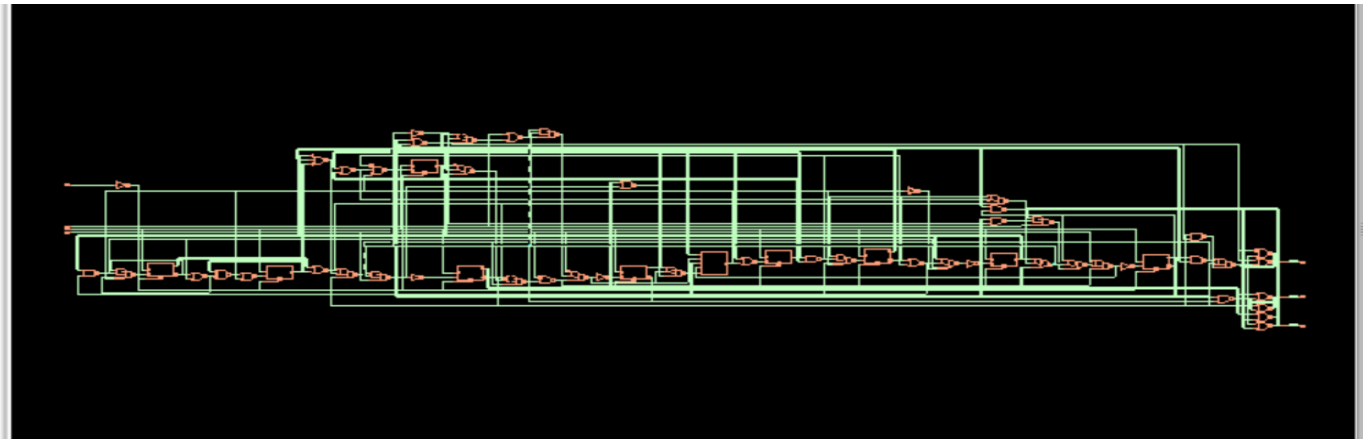


Logic Synthesis Using Cadence Genus

Logic synthesis is the process of converting the RTL description into an optimized gate-level netlist using a standard cell library. In this project, Cadence Genus Synthesis Solution was used to synthesize the Verilog RTL design of the Smart 3-Way Traffic Light Controller.

1.Synthesized Gate-Level Schematic

The RTL design was synthesized using Cadence Genus and mapped into standard cells. The generated gate-level schematic shows the interconnection of logic gates and sequential elements implementing the Smart 3-Way Traffic Light Controller.



2. Area Report

The area report generated after synthesis indicates that the design occupies a total cell area of **389.047 μm^2** and consists of **60 standard cells**.

Open  traffic_area.rpt
~/adithyan/traffic_controller Save  -  

=====
Generated by: Genus(TM) Synthesis Solution 20.11-s111_1
Generated on: Jun 12 2026 03:45:21 pm
Module: smart_traffic_3way
Technology library: slow
Operating conditions: slow (balanced_tree)
Wireload mode: enclosed
Area mode: timing library
=====

Instance	Module	Cell Count	Cell Area	Net Area	Total Area	Wireload
smart_traffic_3way		60	389.047	0.000	389.047	<none> (D)

(D) = wireload is default in technology library

3. Timing Report

The timing analysis report shows a positive setup slack of **7904 ps**, confirming that the synthesized design satisfies the specified timing constraints.

```
traffic_timing.rpt
~/adithyan/traffic_controller

Generated by:      Genus(TM) Synthesis Solution 20.11-s111_1
Generated on:      Jun 12 2026  03:45:21 pm
Module:            smart_traffic_3way
Operating conditions: slow (balanced_tree)
Wireload mode:     enclosed
Area mode:         timing library

=====
Path 1: MET (7904 ps) Setup Check with Pin count_reg[3]/CK->D
  Group: clk
  Startpoint: (R) count_reg[5]/CK
  Clock: (R) clk
  Endpoint: (R) count_reg[3]/D
  Clock: (R) clk

  Capture          Launch
  Clock Edge: +    10000          0
  Src Latency: +    0              0
  Net Latency: +    0 (I)         0 (I)
  Arrival: =       10000          0

  Setup: -         150
  Required Time: =  9850
  Launch Clock: -   0
  Data Path: -     1947
  Slack: =         7904

#-----
```

4. Critical Timing Path

The critical timing path report identifies the longest delay path in the design and verifies that the path meets the required timing specifications without any violations.

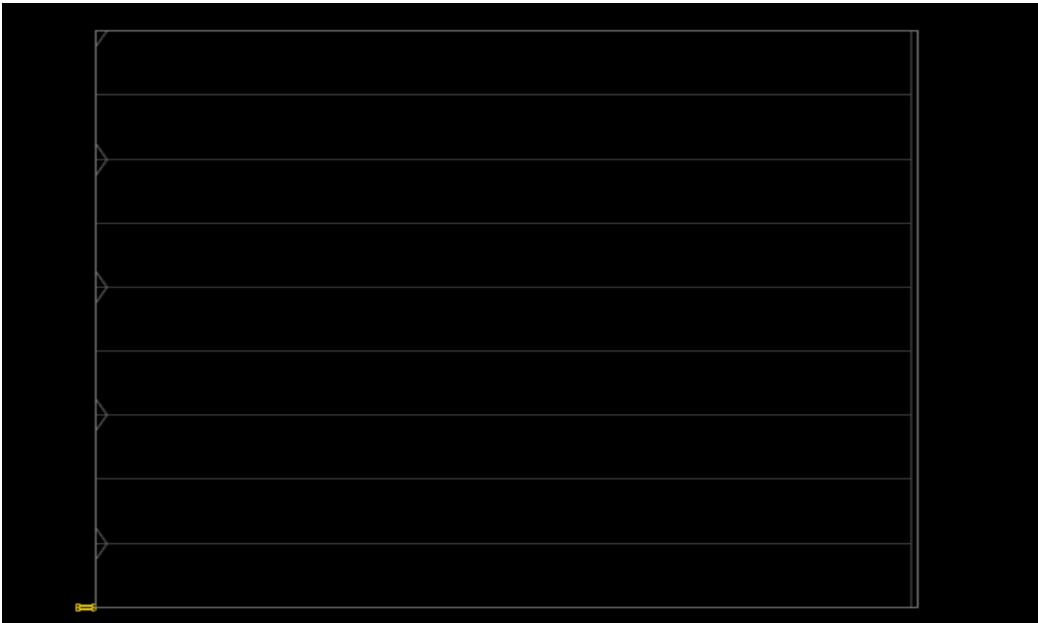
#	Timing Point	Flags	Arc	Edge	Cell	Fanout	Load (fF)	Trans (ps)	Delay (ps)	Arrival (ps)	Instance Location
#	count_reg[5]/CK	-	-	R	(arrival)	9	-	0	0	0	(-, -)
#	count_reg[5]/Q	-	CK->Q	F	DFFRHQX1	3	6.3	67	349	349	(-, -)
	g1920_7098/Y	-	B->Y	R	NOR2XL	3	8.2	263	210	558	(-, -)
	g1904_5477/Y	-	B0->Y	F	OAI21X1	1	2.7	118	111	669	(-, -)
	g1901_9315/Y	-	A1->Y	R	OAI21X1	4	11.1	220	194	864	(-, -)
	g1899/Y	-	A->Y	F	CLKINX1	2	4.5	87	71	935	(-, -)
	g1895_5115/Y	-	A1->Y	R	OAI21X1	1	1.7	79	82	1017	(-, -)
	g1892_7098/Y	-	B->Y	F	NOR2XL	1	2.8	54	58	1076	(-, -)
	g1886_1705/Y	-	B0->Y	R	OAI21X1	1	2.7	88	79	1154	(-, -)
	g1883_1617/Y	-	A1->Y	F	OAI21X1	3	7.1	131	116	1270	(-, -)
	g1878_3680/Y	-	A1->Y	R	OAI21X1	1	2.8	87	104	1374	(-, -)
	g1877_6783/Y	-	C0->Y	F	OAI221X1	6	13.5	358	278	1651	(-, -)
	g1876/Y	-	A->Y	R	INVXL	1	1.7	109	98	1749	(-, -)
	g1872_2398/Y	-	C0->Y	F	OAI211XL	1	1.6	130	122	1871	(-, -)
	g1869/Y	-	A->Y	R	INVXL	1	2.3	67	76	1947	(-, -)
#	count_reg[3]/D	<<<	-	R	DFFRHQX1	1	-	-	0	1947	(-, -)
#	-----										

Physical Design Using Cadence Innovus

1. Floorplanning

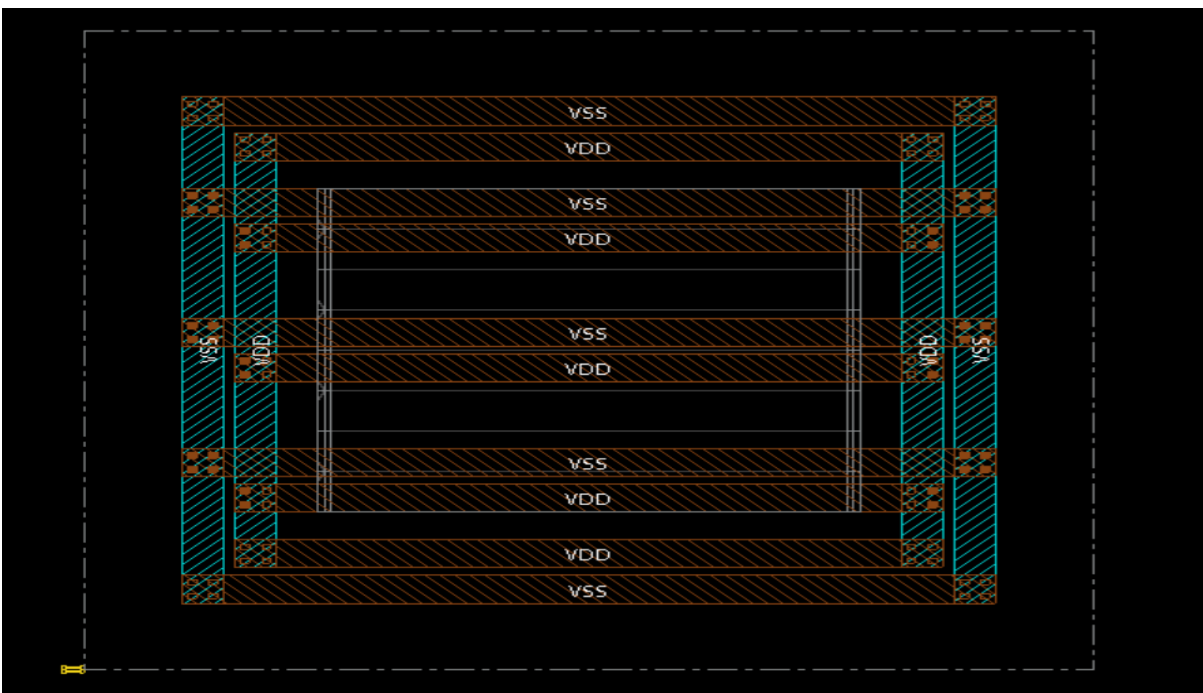
The floorplan shows the core area allocated for the Smart 3-Way Traffic Light Controller along with the placement rows used for standard cell implementation. The floorplan serves as the foundation for subsequent physical design stages.

The floorplan was created in Cadence Innovus with predefined core dimensions and utilization to accommodate the synthesized standard cells.



2. Power Planning

The power planning layout shows the VDD and VSS power rings surrounding the core area along with power stripes used for distributing power and ground across the design. The power network ensures proper connectivity between the power source and standard cells during operation.

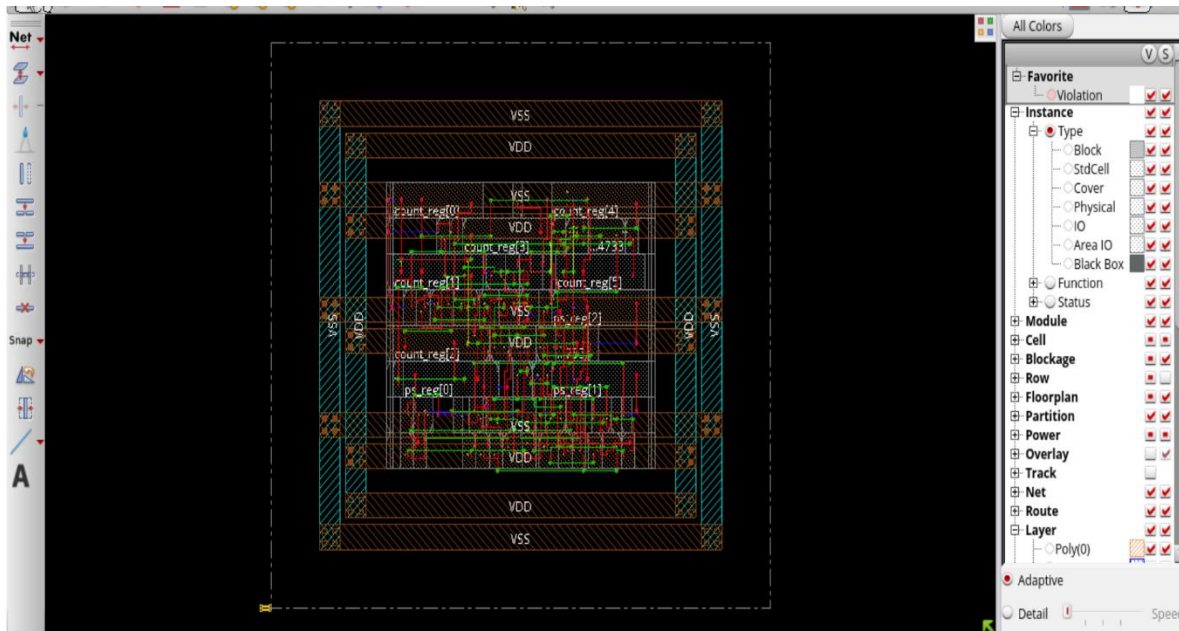


Instance name.

Click to de/select multiple objects.

3. Placement and Routing

After power planning, standard cells were placed within the core region using Cadence Innovus. The placement process optimized cell locations to reduce wire length and improve timing performance. Routing was then performed to establish all signal connections between the placed cells using available metal layers. The final routed design achieved successful connectivity with 0 DRC violations and 0% routing overflow.



4. Timing Analysis

a. Post-Route Setup Timing Analysis

timeDesign Summary				
Setup views included: worst_case				
Setup mode	all	reg2reg	default	
WNS (ns):	7.700	7.700	7.836	
TNS (ns):	0.000	0.000	0.000	
Violating Paths:	0	0	0	
All Paths:	27	9	27	

b. Post-Route Hold Timing Analysis

timeDesign Summary				
Hold views included: best_case				
Hold mode	all	reg2reg	default	
WNS (ns):	0.329	0.329	0.984	
TNS (ns):	0.000	0.000	0.000	
Violating Paths:	0	0	0	
All Paths:	27	9	27	
Density: 83.442%				
Routing Overflow: 0.00% H and 0.00% V				

Table

Timing Summary Table		
Parameter	Setup Analysis	Hold Analysis
WNS (ns)	7.700	0.329
TNS (ns)	0.000	0.000
Violating Paths	0	0
Routing Overflow	0%	0%

5 Power Analysis

Total Power						

Total Internal Power:		0.02197323		80.5212%		
Total Switching Power:		0.00354888		13.0049%		
Total Leakage Power:		0.00176663		6.4738%		
Total Power:		0.02728874				

Group		Internal Power	Switching Power	Leakage Power	Total Power	Percentage (%)

Sequential		0.0175	0.0009919	0.001031	0.01952	71.52
Macro		0	0	0	0	0
I/O		0	0	0	0	0
Combinational		0.004478	0.002557	0.000736	0.007771	28.48
Clock (Combinational)		0	0	0	0	0
Clock (Sequential)		0	0	0	0	0

Total		0.02197	0.003549	0.001767	0.02729	100

Rail	Voltage	Internal Power	Switching Power	Leakage Power	Total Power	Percentage (%)

VDD	0.9	0.02197	0.003549	0.001767	0.02729	100

TABLE

Power Component	Power (mW)	Percentage (%)
Internal Power	0.02197	80.52
Switching Power	0.00355	13.00
Leakage Power	0.00177	6.47
Total Power	0.02729	100

6. Results Summary

Parameter	Result
Design Name	Smart 3-Way Traffic Light Controller
Technology Node	90 nm
Cell Count	60
Synthesis Area	389.047 μm^2
Setup WNS	7.643 ns
Setup TNS	0 ns
Hold WNS	0.339 ns
Hold TNS	0 ns
Setup Violations	0
Hold Violations	0
DRC Violations	0
Routing Overflow	0%
Total Vias	342
Total Power	0.02729 mW
Supply Voltage	0.9 V

Conclusion

This project successfully implemented a Smart 3-Way Traffic Light Controller using the complete RTL-to-GDSII ASIC design flow with Cadence EDA tools. The design was first described in Verilog HDL and functionally verified through simulation. Synthesis was performed using Cadence Genus, followed by floorplanning, power planning, placement, clock tree synthesis, routing, timing analysis, and power analysis using Cadence Innovus.

The final design achieved successful timing closure with positive setup and hold slack values, indicating reliable operation under the specified constraints. Routing was completed with 0 DRC violations and 0% routing overflow, demonstrating a clean physical implementation. Power analysis showed a total power consumption of 0.02729 mW at a supply voltage of 0.9 V, confirming the efficiency of the design.

The project provided practical exposure to the complete ASIC implementation flow and demonstrated how a digital design can be transformed from RTL description to a manufacturable physical layout. The knowledge gained through this project forms a strong foundation for future work in VLSI design, ASIC development, and advanced digital system implementation.